AUTOMATED PLANETARY PERCEPTION SYSTEM

Part 16: Kinodynamic State-Lattice & NMPC Path Optimization

Level 3, Volume 4 of 5: State-Lattice Graphs, Multibody Suspension Constraints, Nonlinear MPC & Safe Stop Arcs

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Document Identifier: APPS-SPEC-L3-VOL4-P16
Milestone Tier: Level 3 Research Prototype Specification (Volume 4 of 5)
Standard Mapping: NASA SP-2016-6105 / NPR 7150.2B Class B Architecture Reference
Release Status: Draft Specification for Peer Review & Simulation Modeling
Total Allocated Pages: 37 Pages (Strict Volume Allocation)

VOLUME SUMMARY: *Specifies hierarchical multi-layer path planning (Topological Global to 3D Kinodynamic Local), 3D state-lattice generation obeying multibody constraints, nonlinear Model Predictive Control (NMPC) trajectory optimization, and fail-safe stopping trajectories.*

Chapter 280. Executive Summary & Kinodynamic Planning Framework

Context & Operational Boundaries: Defines the Level 3 multi-layer trajectory generation and nonlinear predictive control architecture.

- **Hierarchical Planning:** Fast marching global topological guidance combined with local 3D kinodynamic state-lattice trajectory rollouts.
- **Nonlinear MPC Optimizer:** Direct multiple shooting solver optimizing control inputs over a continuous receding time horizon.
- **Fail-Safe Guarantee:** Every commanded trajectory step must possess a certified, collision-free emergency stopping trajectory (ICS avoidance).

r	Algorithm / Model	Prediction Horizon	Execution Rate
	Fast Marching / Topological PRM*	50.0 m to Goal	0.2 Hz (5.0s)
	3D State-Space Kinodynamic Lattice	8.0 m Receding	2.0 Hz (500 ms)
C	Direct Multiple Shooting NLP	3.0 s Dynamic	10.0 Hz (100 ms)
ing	Guaranteed Reachable Safe Stopping Profile	1.5 s Braking Arc	Synchronous (Every Step)

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Chapter 280. Chapter 280.1: Kinodynamic & NMPC Specification 1

Level 3 kinodynamic trajectory optimization specification page 4.

Equation (284.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 4.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 280. Chapter 280.2: Kinodynamic & NMPC Specification 2

Level 3 kinodynamic trajectory optimization specification page 5.

Equation (285.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 5.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 280. Chapter 280.3: Kinodynamic & NMPC Specification 3

Level 3 kinodynamic trajectory optimization specification page 6.

Equation (286.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 6.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 281. Chapter 281.1: Kinodynamic & NMPC Specification 4

Level 3 kinodynamic trajectory optimization specification page 7.

Equation (287.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 7.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 281. Chapter 281.2: Kinodynamic & NMPC Specification 5

Level 3 kinodynamic trajectory optimization specification page 8.

Equation (288.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 8.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 281. Chapter 281.3: Kinodynamic & NMPC Specification 6

Level 3 kinodynamic trajectory optimization specification page 9.

Equation (289.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 9.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 282. Chapter 282.1: Kinodynamic & NMPC Specification 7

Level 3 kinodynamic trajectory optimization specification page 10.

Equation (290.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 10.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 282. Chapter 282.2: Kinodynamic & NMPC Specification 8

Level 3 kinodynamic trajectory optimization specification page 11.

Equation (291.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 11.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 282. Chapter 282.3: Kinodynamic & NMPC Specification 9

Level 3 kinodynamic trajectory optimization specification page 12.

Equation (292.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 12.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 283. Chapter 283.1: Kinodynamic & NMPC Specification 10

Level 3 kinodynamic trajectory optimization specification page 13.

Equation (293.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 13.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 283. Chapter 283.2: Kinodynamic & NMPC Specification 11

Level 3 kinodynamic trajectory optimization specification page 14.

Equation (294.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 14.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 283. Chapter 283.3: Kinodynamic & NMPC Specification 12

Level 3 kinodynamic trajectory optimization specification page 15.

Equation (295.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 15.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 284. Chapter 284.1: Kinodynamic & NMPC Specification 13

Level 3 kinodynamic trajectory optimization specification page 16.

Equation (296.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 16.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 284. Chapter 284.2: Kinodynamic & NMPC Specification 14

Level 3 kinodynamic trajectory optimization specification page 17.

Equation (297.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 17.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 284. Chapter 284.3: Kinodynamic & NMPC Specification 15

Level 3 kinodynamic trajectory optimization specification page 18.

Equation (298.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 18.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 285. Chapter 285.1: Kinodynamic & NMPC Specification 16

Level 3 kinodynamic trajectory optimization specification page 19.

Equation (299.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 19.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 285. Chapter 285.2: Kinodynamic & NMPC Specification 17

Level 3 kinodynamic trajectory optimization specification page 20.

Equation (300.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 20.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 285. Chapter 285.3: Kinodynamic & NMPC Specification 18

Level 3 kinodynamic trajectory optimization specification page 21.

Equation (301.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 21.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 286. Chapter 286.1: Kinodynamic & NMPC Specification 19

Level 3 kinodynamic trajectory optimization specification page 22.

Equation (302.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 22.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 286. Chapter 286.2: Kinodynamic & NMPC Specification 20

Level 3 kinodynamic trajectory optimization specification page 23.

Equation (303.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 23.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

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Level 3 kinodynamic trajectory optimization specification page 24.

Equation (304.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 24.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 287. Chapter 287.1: Kinodynamic & NMPC Specification 22

Level 3 kinodynamic trajectory optimization specification page 25.

Equation (305.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 25.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 287. Chapter 287.2: Kinodynamic & NMPC Specification 23

Level 3 kinodynamic trajectory optimization specification page 26.

Equation (306.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 26.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 287. Chapter 287.3: Kinodynamic & NMPC Specification 24

Level 3 kinodynamic trajectory optimization specification page 27.

Equation (307.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 27.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 288. Chapter 288.1: Kinodynamic & NMPC Specification 25

Level 3 kinodynamic trajectory optimization specification page 28.

Equation (308.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 28.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 288. Chapter 288.2: Kinodynamic & NMPC Specification 26

Level 3 kinodynamic trajectory optimization specification page 29.

Equation (309.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 29.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 288. Chapter 288.3: Kinodynamic & NMPC Specification 27

Level 3 kinodynamic trajectory optimization specification page 30.

Equation (310.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 30.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 289. Chapter 289.1: Kinodynamic & NMPC Specification 28

Level 3 kinodynamic trajectory optimization specification page 31.

Equation (311.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 31.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 289. Chapter 289.2: Kinodynamic & NMPC Specification 29

Level 3 kinodynamic trajectory optimization specification page 32.

Equation (312.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 32.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 289. Chapter 289.3: Kinodynamic & NMPC Specification 30

Level 3 kinodynamic trajectory optimization specification page 33.

Equation (313.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 33.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 290. Chapter 290.1: Kinodynamic & NMPC Specification 31

Level 3 kinodynamic trajectory optimization specification page 34.

Equation (314.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 34.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 290. Chapter 290.2: Kinodynamic & NMPC Specification 32

Level 3 kinodynamic trajectory optimization specification page 35.

Equation (315.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 35.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 290. Chapter 290.3: Kinodynamic & NMPC Specification 33

Level 3 kinodynamic trajectory optimization specification page 36.

Equation (316.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 36.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier

Chapter 291. Chapter 291.1: Kinodynamic & NMPC Specification 34

Level 3 kinodynamic trajectory optimization specification page 37.

Equation (317.1): NMPC Objective Function & Risk Constraints

```
J(x, u) = integral_0^T [ w_dist * ||x - x_goal||^2 + w_risk * r(x) + w_slip * s(x)^2 + w_energy * ||u||^2 ] dt
                s.t. dx/dt = f_terramechanics(x, u), FASM(x) >= 15.0 N*rad, ESDF(x) >= R_body
```

Where: J is cost functional, u is control input (torque, steering), r(x) is traversability risk, s(x) is predicted soil slip, FASM is stability margin.

- **Multibody Feasibility Audit:** Verifies that roll, pitch, and wheel torque bounds are strictly respected along primitive 37.
- **Active Perception Gain:** Incentivizes trajectories that maintain feature-rich visual landmarks within camera field of view.

Constraint Type	Physical Bound	Enforcement Method
Max Linear Velocity	0.25 m/s	Hard NLP Box Constraint
Max Curvature Rate	0.50 rad/s^2	Actuator Slew Rate Limiter
Dynamic Rollover Margin	FASM > 15.0 N*rad	Nonlinear Penalty Barrier